

DSP Lab-4

Realization of Discrete Time Systems

1. PROPERTIES OF SYSTEMS:

1.1 CAUSAL AND NON-CAUSAL SYSTEMS:

A system is causal if the system output $y[n_0]$ at time $n=n_0$ does not depend on values of the input $x[n]$ for $n > n_0$. In other words, for any input signal $x[n]$, the corresponding output $y[n]$ depends upon the present and past values of $x[n]$. So, if the input to a causal system is zero for $n < n_0$, the output of the system is also zero for $n < n_0$. All natural systems are causal.

1.2 STATIC (MEMORY LESS) AND DYNAMIC (WITH MEMORY) SYSTEMS:

A system is static or memory less if for any input signal $x(t)$ or $x[n]$ the corresponding output $y(t)$ or $y[n]$ depends only on the value of the input signal at that time. A non-static system is called dynamic or dynamical.

1.3 LINEAR AND NON-LINEAR SYSTEMS:

Let $y(t)$ denote the response of the system S to an input signal $x[n]$, that is, $y[n] = S\{x[n]\}$. System S is linear if for any input signal $x_1[n]$ and $x_2[n]$ and any scalar a_1 and a_2 the following relationship equation holds.

$$S\{a_1 x_1[n] + a_2 x_2[n]\} = a_1 S\{x_1[n]\} + a_2 S\{x_2[n]\}$$

In other words, the response of the linear system to an input that is a linear combination of two signals is the linear combination of the responses of the system to each one of these signals. The linearity property is generalized for any number of input signals, and this is often referred to as the principle of superposition. The linearity property is the combination of two other properties: the additive property and the homogeneity property. A system S satisfies the additivity property if for any input signals $x_1[n]$ and $x_2[n]$

$$S\{x_1[n] + x_2[n]\} = S\{x_1[n]\} + S\{x_2[n]\}$$

While the homogeneity property implies that for any scalar a and any input signal $x(t)$,

$$S\{ax[n]\} = aS\{x[n]\}$$

1.4 STABLE AND UNSTABLE SYSTEMS:

Stability is very important system property. The practical meaning of a stable system is that for a small applied input the system response is also small (does not diverge). A more formal definition is that a system is stable or bounded-input bounded-output (BIBO) stable if the system response to any bounded-input signal is bounded-output signal. The mathematical expression is as follows:

Suppose that a positive number $M < \infty$ exists, such that $|x[n]| \leq M$. The system is stable if a positive number $N < \infty$ exists, such that $|y[n]| \leq N$. A non-stable system is called unstable system.

1.5 TIME-INVARIANT AND TIME-VARIANT SYSTEMS:

A system is time invariant, if a time shift in the input signal results in the same time shift in the output signal. In other words, if $y[n]$ is the response of a time-invariant system to an input signal $x[n]$, then the system response to the input signal $x(n - n_o)$ is $y(n - n_o)$. The mathematical expression

$$y(n - n_o) = S x(n - n_o)$$

From the above equations, we conclude that if a system is time invariant, the amplitude of the output signal is the same independent of the time instance the input is applied. The difference is time shift in the signal input. A non-time invariant system is called time-varying or time-variant system.

1.6 Differentiator

In **continuous time**, differentiation means:

$$y(t) = \frac{d}{dt}x(t)$$

In **discrete time**, we approximate the derivative by the **first difference**: $y[n] = x[n] - x[n - 1]$. This is also called a **differentiator**.

- It highlights **changes** in the signal (edges, transitions).
- If the signal is constant, output = 0.

1.7 Accumulator

In **continuous time**, integration means:

$$y(t) = \int x(t) dt$$

In **discrete time**, the accumulator is the running sum:

$$\begin{aligned} y(n) &= \sum_{k=-\infty}^n x(k) = \sum_{k=-\infty}^{n-1} x(k) + x(n) \\ &= y(n-1) + x(n) \end{aligned}$$

It's also called the **discrete-time integrator**.

- It keeps adding input values over time.
- Useful for signal smoothing or as part of IIR filters.

In Lab Task

In-Lab Task 01: Find if discrete-time system described by the I/O relationship $y[n] = -x[n]$

- a) Static or Dynamic (input signal $x[n] = 2n, -2 \leq n \leq 2$)
- b) Causal or non-causal (input signal $x[n] = 2n, -2 \leq n \leq 2$)
- c) Linear or non-linear (input signals $x_1[n] = 2n, -2 \leq n \leq 2, x_2[n] = n^3, -2 \leq n \leq 2, a_1 = 2$ and $a_2 = 3$)
- d) Shift invariant or shift variant (input signal $x[n] = 2n, -2 \leq n \leq 2$ and shift $n_0 = 3$)

In-Lab Task 02: Consider the discrete-time signal and implement a differentiator and accumulator

$$x(n) = \begin{cases} |n|, & -3 \leq n \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

- a. Plot the original signal **x[n]**, the differentiated signal **y[n]**, and the accumulated signal **z[n]** using **stem plots**.
- b. Comment on the results:
 - How does the differentiator respond to changes in the signal?
 - How does the accumulator modify the signal?

In-Lab Task 03: Given the discrete-time signal

$$x(n) = \{2, 4, 6, 8, 10\}, \quad 0 \leq n \leq 4$$

- Plot the original signal using a stem plot.
- Downsample and Upsample the signal by a factor of 2.
- Plot and compare both signals on the same axis.

Post Lab Task

Post-Lab Task 1: Find if discrete-time system described by I/O relationship $y[n] = x[1-2n]$ is:

- Static or Dynamic (input signal $x[n] = 2n, -2 \leq n \leq 2$)
- Causal or non-causal (input signal $x[n] = 2n, -2 \leq n \leq 2$)
- Linear or non-linear (input signals $x_1[n] = 2n, -2 \leq n \leq 4, x_2[n] = n/3, -2 \leq n \leq 4, a_1 = 2$ and $a_2 = 3$)
- Shift invariant or shift variant (input signal $x[n] = 2n, -2 \leq n \leq 4$ and shift $n_0 = 3$)

Post-Lab Task 01: Consider the system: $y[n] = y[n-1] + x[n], \quad y[-1] = 0$

With input (unit step): $x[n] = u[n] - u[n-10]$ (a 10-point rectangular pulse)

- Implement this accumulator system in MATLAB.
- Compare your results.
- Comment on how the accumulator behaves as a discrete integrator as an averaging filter.